Yassa Ashraf

Software Engineer | Backend, Cloud & AI Systems

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PROFESSIONAL SUMMARY

Software Engineer specializing in backend, cloud, and AI-enabled systems across enterprise and product environments. Experienced in TypeScript, Node.js, NestJS, Java, Spring Boot, ASP.NET, React, Next.js, and AWS, with hands-on delivery across APIs, microservices, event-driven systems, infrastructure as code, CI/CD, and production operations. Strong background in scalable system design, cloud-native development, application reliability, and integrating AI capabilities into real-world software products.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C#, SQL, Dart, Bash
Backend: Node.js, NestJS, Express, Spring Boot, ASP.NET Web API, REST, GraphQL, Microservices
Frontend & Mobile: React, Next.js, Angular, React Native, Flutter, Tailwind CSS
Cloud & DevOps: AWS, EC2, Lambda, ECS, EKS, S3, API Gateway, CloudFront, CloudWatch, SageMaker, Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD
Data: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Prisma, Mongoose, JDBC
AI & ML: Generative AI, LLMs, RAG, Agentic AI, Multi-Agent Systems, Hugging Face Transformers, ChemBERTa, Gemini, Google ADK, Embeddings, Semantic Retrieval, Explainable AI
Architecture & Security: Clean Architecture, Domain-Driven Design, Event-Driven Architecture, Serverless Architecture, Distributed Systems, OAuth2, JWT, RBAC, API Security
Testing & Observability: JUnit, Jest, Unit Testing, Integration Testing, CloudWatch, Grafana, Logging, Monitoring

PROFESSIONAL EXPERIENCE

Mid-Level Software Engineer — Deloitte Cairo, Egypt (Hybrid)
Sep 2026 – Present | Full-time

- Design and build cloud-native backend services and platform components using AWS, TypeScript, Node.js, Terraform, and modern software engineering practices.
- Develop secure APIs, event-driven workflows, and infrastructure components for enterprise platforms operating in regulated environments.
- Implement infrastructure as code and CI/CD workflows supporting repeatable validation, deployment, and environment consistency.
- Collaborate with cross-functional and international teams on cloud architecture, integration, testing, security, and production delivery.

Senior Software Engineer — EV Cozmo Greater London, UK (Remote)
Jul 2026 – Sep 2026 | Contract

- Reverse-engineered mobile clients, local data stores, and network behavior across 10+ EV charging applications to integrate station and live-availability data where public APIs were incomplete or unavailable.
- Recovered and normalized station inventories from Firestore caches, local application storage, and network responses, significantly increasing charging-location coverage across integrated networks.
- Architected a pluggable live-status provider layer synchronizing multiple charging networks in parallel with source-level timeouts, rate-limit-aware authentication, and fail-soft error handling.
- Built geospatial entity-resolution and deduplication pipelines using MongoDB geospatial queries and GeoJSON indexing to map third-party charging records to canonical stations.
- Developed clean-architecture station services for CRUD, nearest-station search, favorites, and image management using AWS Lambda, API Gateway, and Mongoose.
- Delivered Flutter station-discovery features including maps, filters, favorites, and production releases to the Apple App Store and Google Play.
- Maintained CI/CD pipelines with GitHub Actions and SST across development and production environments.

Associate Software Engineer — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Jul 2025 – Sep 2026 | Full-time

- Developed enterprise applications using ASP.NET Web API, Angular, Entity Framework Core, and reusable backend architecture across SerieTool, VEB Tool, and DLM Pricing Tool.
- Designed and built a reusable ASP.NET Core project template adopted across multiple internal applications, implementing JWT authentication with token revocation, generic repository and Unit of Work patterns, centralized dependency injection, Serilog, Hangfire, AutoMapper, and audit tracking.

- Built SerieTool's Angular admin panel for workforce and order management, including agent management, manual order assignment, backlog handling, order categorization, product management, leave management, and daily reporting.
- Designed and delivered the VEB Tool from initial architecture to production, building an Excel-based order ingestion pipeline integrated with RPA workflows, order lifecycle tracking, delay monitoring, and full audit history.
- Developed analytics dashboards for order statistics, delivery performance, customer insights, and operational bottlenecks, alongside role-based access control and customer management features.
- Contributed to the DLM Pricing Tool by developing backend and frontend functionality, resolving production issues, implementing advanced filtering, and improving enterprise order and pricing workflows.
- Improved code quality and maintainability through shared components, standardized service-layer patterns, stronger validation and error handling, code reviews, and mentoring junior developers.
- Supported UAT, stakeholder demonstrations, release preparation, and production delivery within enterprise security and development standards.

Software Engineer Intern — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Feb 2025 – Jul 2025 | Internship

- Contributed to ASP.NET Web API backend enhancements and Angular frontend development.
- Improved performance of critical endpoints through database query and API optimization.
- Implemented unit tests and participated in peer reviews and software quality processes.

AI Software Engineer — VIXI AI Remote, UAE
Sep 2025 – Jan 2026 | Contract

- Designed NLP and retrieval pipelines using transformer-based models for enterprise AI use cases.
- Built containerized AI inference services with deployment automation, autoscaling, and monitoring.
- Collaborated with product and data teams to translate business requirements into production AI capabilities.
- Improved development velocity by introducing preview and staging environments for faster iteration.

Software Engineer — Xylem Software Remote, Egypt
Jun 2025 – Nov 2025 | Contract

- Delivered web and mobile functionality in a monorepo using React, React Native, Node.js, and shared application services.
- Led backend development for analytics and operational platforms, including ingestion APIs and dashboard services.
- Improved application performance through caching, query optimization, and backend service improvements.

Mid-Level Backend Engineer — Trevi Cairo, Egypt
Oct 2024 – Sep 2025 | Full-time

- Architected wallet and loyalty backend services using NestJS, Prisma, and relational databases.
- Integrated cashback and promotion workflows supporting transaction-heavy financial use cases.
- Deployed services on AWS with health checks, retry mechanisms, monitoring, and reliability controls.

Software Engineer — MM Solutions Remote, USA
Jul 2024 – Mar 2025 | Part-time

- Led web integration work and optimized frontend loading, caching, and application performance.
- Automated dispatch and operational workflows supporting Amazon Freight partner operations.

Full Stack Developer — Kinetics Egypt Cairo, Egypt
Aug 2024 – Oct 2024

- Delivered cloud-hosted software solutions using AWS EC2 and S3 with secure environment configuration.
- Developed product features in collaboration with engineering and product teams under tight delivery timelines.

Full Stack Developer — Fiverr Remote
May 2023 – Oct 2023 | Freelance

- Built custom web applications, dashboards, and interactive products for international clients.
- Managed delivery from requirements gathering and implementation through deployment and handover.

Full Stack Developer Intern — Instatus Cairo, Egypt
Jul 2022 – Sep 2022 | Internship

- Developed frontend functionality using Next.js, GraphQL, and Tailwind CSS.
- Integrated GraphQL queries and mutations for real-time status-management workflows.
- Improved UI responsiveness and collaborated with backend engineers on schema integration.

SELECTED PROJECTS

Beity App — Cloud Kitchens Ordering and Delivery System NestJS, React Native, AWS, PostgreSQL

- Built a cloud-kitchen platform supporting ordering, delivery management, users, and operational workflows.
- Implemented order lifecycle management, role-based authorization, clean backend architecture, and AWS deployment pipelines.

Intervyo — Smart Interview Platform Next.js, Node.js, TypeScript, PostgreSQL

- Developed an interview-management platform supporting technical and HR workflows, scheduling, candidate evaluation, and REST APIs.

E-Commerce Platform for Plastic Pallets NestJS, Next.js, Kafka

- Designed product, order, and payment microservices using Kafka for asynchronous communication and resilient service integration.

Bookstore Serverless Application AWS Lambda, API Gateway, DynamoDB

- Built serverless CRUD and search APIs with authentication, rate limiting, DynamoDB access patterns, CloudWatch monitoring, and CI/CD.

ECS Help Desk System MERN Stack, Tailwind CSS

- Built a ticket-management platform supporting roles, SLAs, real-time updates, analytics, and operational workflows.

EDUCATION

Bachelor of Computer Science — Software Engineering Oct 2021 – Jul 2025
German International University, Cairo

RESEARCH & PUBLICATIONS

Bachelor Thesis — Using Explainable AI Algorithms for Material Recognition Grade: A+

- Built ChemBERTa-based material-recognition models with post-hoc explainability techniques.
- Enabled traceable predictions using attention visualization and feature-attribution methods.
- Deployed Flask inference services using AWS SageMaker, EC2, and S3 with CI/CD pipelines.
- Delivered a Next.js dashboard for real-time inspection of model predictions and explanations.

MoleculeX-Predictive: An augmentation of transformers with explainable AI and knowledge layers for efficient drug property prediction
Yassa Ashraf, Daniel Sameh, John Zaki, Amal Aboulhassan
Array (Elsevier), 2026
DOI: 10.1016/j.array.2026.100820 | ScienceDirect

CERTIFICATIONS

Explainable AI (XAI) Specialization — Duke University Issued Oct 2025
Verify Specialization | 3 Courses
Explainable Machine Learning (XAI) — Duke University Issued Oct 2025
Verify Credential
Interpretable Machine Learning — Duke University Issued Oct 2025
Verify Credential
Oracle Cloud Infrastructure Foundations Associate (2025) Issued Mar 2025
Verify Credential
Career Essentials in Generative AI — Microsoft / LinkedIn Issued Oct 2025
LinkedIn Profile
Ethics in the Age of Generative AI — LinkedIn Issued Oct 2025
LinkedIn Profile

LANGUAGES

English: Professional Proficiency | **Arabic:** Native Proficiency | **German:** Limited Working Proficiency